

MovieLens Dataset Results

(a) RS domain					(b) IR domain				
Approach	@k	NDCG	Precision	Recall	Approaches	@k	NDCG	Precision	Recall
RS models					IR models				
LightFM	5	0.6467 [†]	0.7646 [†]	0.2719	BM25	5	0.0641	0.0597	0.0066
	10	0.6481 [†]	0.6887 [†]	0.4774		10	0.0555	0.0460	0.0099
	20	0.6701 [†]	0.5273	0.6937		20	0.0463	0.0352	0.0142
	50	0.7471	0.2643	0.8491		50	0.0390	0.0246	0.0202
DeepFM	5	0.5490 [†]	0.6375 [†]	0.2450	Dense Retriever	5	0.0697	0.0692	0.0021
	10	0.5738 [†]	0.5970 [†]	0.4459		10	0.0734	0.0731	0.0042
	20	0.6326 [†]	0.4922 [†]	0.6955		20	0.0745	0.0762	0.0082
	50	0.7354 [†]	0.2684 [†]	0.9068		50	0.0770	0.0772	0.0198
BERT4Rec	5	0.6753	0.7969	0.2890	Bi-Encoders	5	0.0845	0.0780	0.0097
	10	0.6955	0.7426	0.5253		10	0.0771	0.0696	0.0160
	20	0.7441	0.5957	0.7937		20	0.0700	0.0602	0.0255
	50	0.8297	0.2992	0.9654		50	0.0671	0.0498 [†]	0.0464
LightGCN	5	0.6080	0.6932	0.2919	KNN-SPLADE	5	0.0597	0.0551	0.0085
	10	0.6330	0.6306	0.5096		10	0.0541	0.0482	0.0138
	20	0.6909	0.5046	0.7618		20	0.0524	0.0454	0.0224
	50	0.7871	0.2679	0.9631		50	0.0513	0.0360	0.0398
ItemKNN	5	0.4375	0.5018	0.2602	ColBERT	5	0.1248	0.1123	0.0142
	10	0.5420	0.5311	0.5257		10	0.1183	0.1033	0.0210
	20	0.6422	0.4582	0.8062		20	0.1093	0.0913	0.0351
	50	0.7159	0.2368	0.9702		50	0.0983	0.0753	0.0610
RS → IR mapping					IR → RS mapping				
Centroid Vector ^a	5	0.3786	0.3654	0.1563	Query-as-User ^c	5	0.0834	0.0806	0.0265
	10	0.3738	0.3259	0.2682		10	0.0828	0.0761	0.0392 [†]
	20	0.4013	0.2766	0.4363		20	0.0845	0.0706	0.0609 [†]
	50	0.5564	0.2018	0.7593		50	0.1027	0.0617 [†]	0.1037 [†]
Tag Query ^b	5	0.5413	0.5209	0.2157	Retrieval-as-User ^c	5	0.0847	0.0819	0.0232
	10	0.5071	0.4336	0.3440		10	0.0813	0.0763	0.0428
	20	0.4790	0.3080	0.4672		20	0.0778	0.0640	0.0723
	50	0.5475	0.1643	0.6054		50	0.1040	0.0452 [†]	0.1299

Table 1: MovieLens: BR2IDGE top-k mean results. **Bold** marks best values per @k/metric; [†] indicates statistical equivalence (t-test, p-mode=holm, $\alpha = 0.05$). Hybrid backbones: ^a Centroid Vector uses *Bi-Encoder*; ^b Tag Query uses *BM25*; ^c IR → RS strategies use *LightFM*.